Tonmoy Saha Choyon

MSc Data Science Graduate, University of Aberdeen
Aberdeen, UK | choyontonmoysaha@gmail.com | +44 7562 544303 | ORCID: 0009-0000-5459-6872
linkedin.com/in/tonmoysaha12 | github.com/tonmoysahachoyon | tonmoysahachoyon.github.io

RESEARCH SUMMARY

My work centers on generative AI and the validation problem underneath it, establishing how far a model's output can be trusted when little or no real ground truth exists to check it against. My MSc dissertation, conducted in collaboration with XYNQ (UK), addressed a case where no real training data was available, building a synthetic data pipeline grounded in limited market data and domain assumptions, and statistically validating it end to end. I am seeking a PhD position in machine learning focused on trustworthy generative models and the rigorous evaluation of synthetic data.

EDUCATION

University of Aberdeen, Aberdeen, UK — January 2025 – June 2026
Master of Science in Data Science — Commendation
Dissertation: *Synthetic Data Generation and Model Validation* (in partnership with XYNQ)
Supervisor: Prof. M. Carmen Romano
Relevant modules: Machine Learning, Advanced Statistics and Special Applications, Image Analysis, Data Visualization, Introduction to Python and R

American International University-Bangladesh, Dhaka, Bangladesh — September 2019 – May 2023
Bachelor of Science in Computer Science and Engineering — Cumulative GPA: 3.99/4.00
Thesis: Compared three classification approaches (CNN, SVM, and an ensemble classifier) for medical image classification, analysing failure modes across approaches rather than optimising a single model in isolation.
Relevant modules: Artificial Intelligence, Algorithms, Data Structures

HONOURS AND AWARDS

- **Academic Excellence Scholarship**, American International University-Bangladesh
- **Summa Cum Laude**, American International University-Bangladesh — 2023
- **Dean's List**, American International University-Bangladesh (four semesters)

RESEARCH AND TECHNICAL PROJECTS

Synthetic Data Generation and Model Validation (MSc Dissertation, with XYNQ) — Aberdeen, UK

- Addressed fraud detection in music streaming where no real labelled training data was available, by constructing a baseline dataset from prompt-engineered LLM outputs grounded in limited available market data and explicit domain assumptions.
- Used CTGAN to expand the baseline into a full synthetic dataset, and tested scalability by training on a 5,000-row subset and generating 100,000 rows, evaluating whether a small seed dataset could be reliably expanded.
- Ran an ablation study across input features to identify those most influential to model performance, using the result to prioritise features in the generation pipeline.
- Measured the fidelity of the generated data to the baseline distribution using four statistical tests covering distinct failure modes: KS distance (marginal distribution shift), Total Variation Distance, Spearman correlation (dependency structure), and Cramér's V (categorical association).
- Trained a Random Forest classifier on the synthetic data and evaluated it on held-out data, reaching an overall F1 score of 0.85 and ROC-AUC scores of 0.96, 0.93, and 0.89 across three fraud categories.
- Identified the central limitation of the approach: fidelity and classifier performance are measured relative to the assumption-grounded baseline rather than real fraud data, motivating validation against real samples or domain-expert review as the next step.

Lead Scoring and Sales Intelligence Pipeline
`github.com/tonmoysahachoyon/leads-score-sales-intelligence`

- Restructured an exploratory notebook into a reproducible two-stage pipeline (train.py, score.py) with shared preprocessing, resolving feature misalignment between training and scoring passes.
- Tuned a Random Forest classifier to 0.95 precision and 0.88 recall, then applied K-Means clustering to separate converted leads into five behavioural segments, isolating a low-intent group that would otherwise have diluted the scoring signal.
- Used SHAP values to attribute each prediction to its dominant feature, converting the model output into a ranked, explainable list rather than an opaque score.

Customer Segmentation via RFM Modelling and K-Means Clustering
`github.com/tonmoysahachoyon/rfm-customer-segmentation`

- Derived Recency, Frequency, and Monetary features from raw transactional data and corrected for right skew using logarithmic scaling before clustering.

- Segmented the customer base into four behavioural groups via K-Means, identifying over 1,000 inactive accounts and a distinct high-value segment.

Job Market Trend Analysis via Cloud Data Warehousing and Power BI
`github.com/tonmoysahachoyon/job-market-pipeline`

- Built an automated extraction pipeline against the Adzuna API, loading raw JSON into Google BigQuery and defining SQL transformations and automated tests using dbt.
- Constructed a Power BI dashboard to track posting volume, hiring activity by company, and salary trends over time.

WORK EXPERIENCE

Family Restaurant Business Dhaka, Bangladesh
Data & Operations Lead January 2024 – November 2024

- Analysed sales and inventory data to inform stocking and scheduling decisions, reducing stock waste by approximately 30 percent.
- Designed and deployed a point-of-sale system to replace manual record-keeping, reducing daily administrative time by roughly two hours.

DeshiIT Dhaka, Bangladesh
Junior Software Developer Intern January 2023 – April 2023

- Restructured a MongoDB schema across more than 10,000 records, reducing query response times by approximately 40 percent.
- Built full-stack features using the MERN stack (MongoDB, Express.js, React, Node.js).

TECHNICAL SKILLS

- **Generative AI and Machine Learning:** CTGAN, synthetic data generation, LLM prompt engineering, scikit-learn, Random Forest, K-Means, SHAP, statistical validation (KS distance, TVD, Spearman, Cramér's V)
- **Programming:** Python, R, JavaScript, Wolfram Mathematica
- **Data Engineering:** SQL, Google BigQuery, dbt, ETL pipelines, API integration
- **Visualisation and Analysis:** Power BI, Tableau, pandas, NumPy
- **Tools:** Git, GitHub, MySQL, MongoDB, Google Cloud CLI

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic: Overall Band 8.0
Listening: 9.0 | Reading: 8.0 | Speaking: 7.5 | Writing: 6.5

REFEREES

Prof. M. Carmen Romano
Professor, School of Natural and Computing Sciences
University of Aberdeen
MSc Dissertation Supervisor
E-mail: m.romano@abdn.ac.uk

Tanvir Ahmed
Assistant Professor, Department of Computer Science
American International University-Bangladesh, Dhaka
Course Instructor, BSc in Computer Science and Engineering
E-mail: tanvir.ahmed@aiub.edu